

1.5 Systems software

Where this unit sits, what it covers, and how far

Where this unit sits

Qualification	GCSE Computer Science (9-1), specification code J277, from 2020
Component	Component 01: Computer systems — 80 marks, 1 hour 30 minutes
What OCR says the component covers	the central processing unit (CPU), computer memory and storage, data representation, wired and wireless networks, network topologies, system security and system software
This unit	topic 1.5 Systems software
Checked	3 October 2026, at https://www.ocr.org.uk/qualifications/gcse/computer-science-j277-from-2020/specification-at-a-glance/

How far this has been checked. The topic numbers and titles below are the ones this course uses on its worksheets and slides. They were not re-checked against the published specification document when this map was built: the document could not be retrieved from this build environment. Check them against <https://www.ocr.org.uk/Images/558027-specification-gcse-computer-science-j277.pdf> before relying on this map for a scheme of work, and treat any finer subdivision as absent rather than assumed - no sub-point numbers are claimed anywhere in these documents, because none were verified.

Before it: 1.1 Systems architecture, 1.2 Memory and storage, 1.3 Computer networks, connections and protocols, 1.4 Network security

After it: 1.6 Ethical, legal, cultural and environmental impacts of digital technology, 2.1 Algorithms, 2.2 Programming fundamentals, 2.3 Producing robust programs, 2.4 Boolean logic, 2.5 Programming languages and Integrated Development Environments, PY Python programming

The lesson sequence and the questions it answers

#	LESSON	THE BIG QUESTION	ACTIVITIES	MARKS
1	Control room ss-101	What does an operating system do, and why does every computer need one?	13	28
2	Multitasking floor ss-102	How does an operating system share out the processor and the memory?	14	22
3	Accounts office ss-103	How does an operating system manage users and their files?	14	26
4	Utility workshop ss-104	What is the purpose of utility software?	14	27
5	End of topic test ss-105-test	assesses the whole unit	0	0

Prior knowledge, vocabulary and skills

Assumed before the unit: the outcomes of 1.1, 1.2, 1.3, 1.4

Vocabulary introduced: Operating system, User interface, Control room, Multitasking, Device driver, Multitasking floor, User account, File management, Accounts office, Utility software, Defragmentation, Utility workshop

What a pupil does, not only reads: defrag, match, mcq, memory, multi, order, permissions, sort

Coverage matrix

Every outcome this unit states, and whether it is taught, practised and assessed **within this unit**. Full means all three; partial means some; not covered means none of the three could be traced.

How this was worked out. The trace is derived by matching the content words of an outcome against the station text, the activities and the exam questions of the unit, not authored by hand. The words each match was made on are in build/coverage.json. It shows where to look; it does not certify that the alignment is right, and a full row has not been checked by a person either.

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
know the purpose and functionality of operating systems ss-l01-o1	ss-l01	yes	yes	yes	Full
know the different types of user interface and the features of each ss-l01-o2	ss-l01	yes	yes	yes	Full
know what is meant by the term multitasking ss-l02-o1	ss-l02	yes	yes	yes	Full
understand how the operating system manages memory ss-l02-o2	ss-l02	yes	yes	yes	Full
understand the need for device drivers ss-l02-o3	ss-l02	yes	yes	yes	Full
understand what is meant by user management ss-l03-o1	ss-l03	yes	yes	yes	Full
understand the ways an operating system manages files ss-l03-o2	ss-l03	yes	yes	yes	Full
understand encryption utilities ss-l04-o1	ss-l04	yes	yes	yes	Full
understand defragmentation utilities ss-l04-o2	ss-l04	yes	yes	yes	Full
understand data compression utilities ss-l04-o3	ss-l04	yes	yes	yes	Full

10 full · 0 partial · 0 not covered, of 10 outcomes.

Gaps, assumptions and overlap

None found by the trace.

Assessed, but matching no outcome this unit states

Each of these is a real exam question on a worksheet in this unit. Either the outcome it tests is worded differently, or it is genuinely assessing something the unit does not claim to teach.

- ss-l03-x2 Explain why a network manager sets permissions on groups rather than on individual users.